

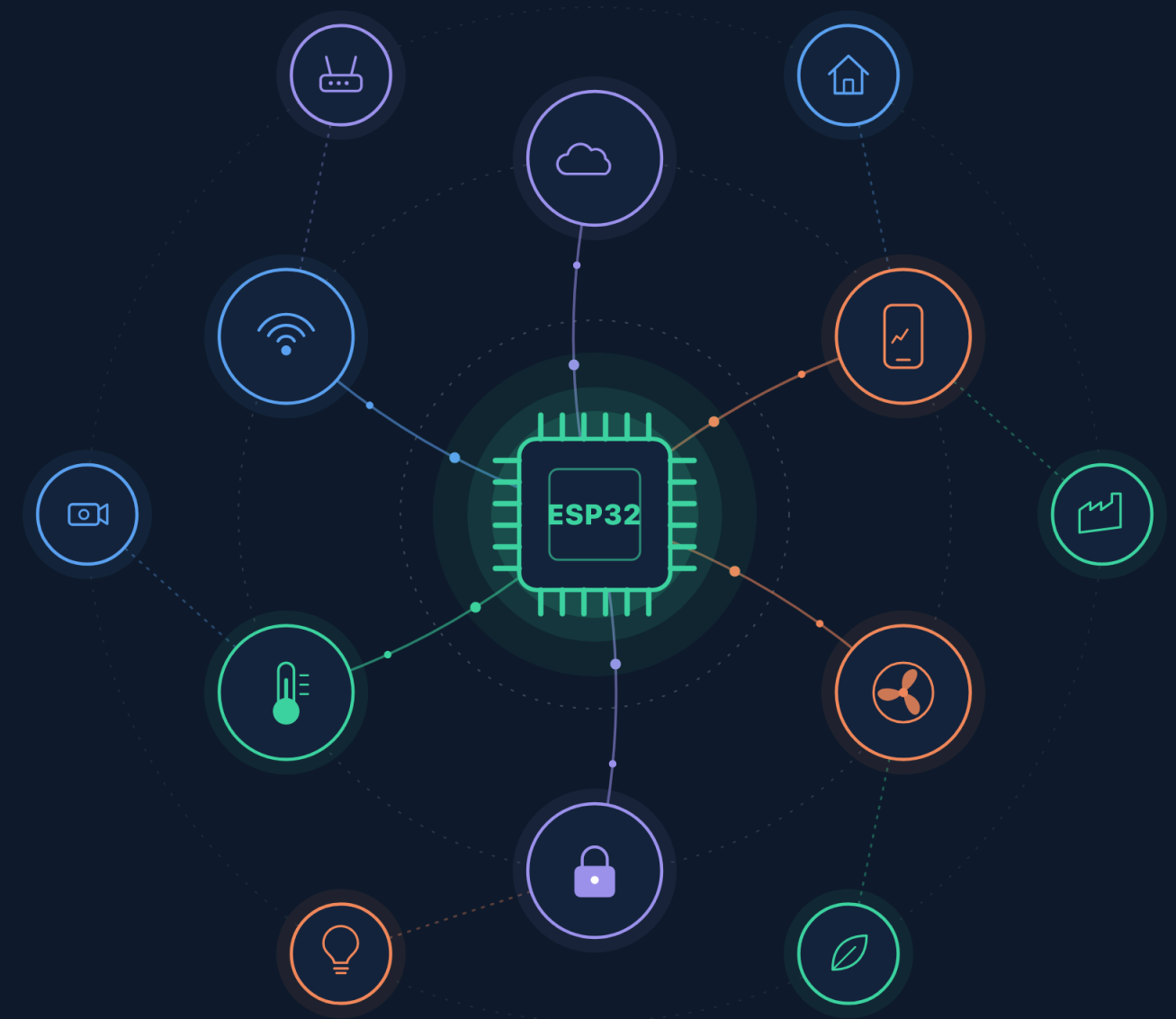
MODULE 8 • 45 MINUTES

Real-World IoT Applications and Project Showcase

How professionals design real IoT systems, how real industries use them, and your own capstone project.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD SO FAR

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

Lecture

02

How it's built

Lecture

03

Sensors and actuators

Labs 1–2

04

Getting online

Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and dashboards

Lab 4

06

Automation

Lab 5

07

Security

Group activity

08

Real world + capstone

You are here

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Follow** a step-by-step method to design an IoT system
 - 02 **Explain** the design choices in homes, industry and farms
 - 03 **Describe** how teams prototype an idea quickly and cheaply
 - 04 **Identify** the main challenges of deploying IoT for real
 - 05 **Build and present** a complete IoT project
-

THE BIG PICTURE

From problem to deployed system



1. Problem

What problem, and for whom?



2. Requirements

What must it do, and where?



3. Design

Choose each of the five layers.



4. Prototype

Simulate first, then real hardware.



5. Pilot

A small test in the real world.



6. Deploy

Run it — then learn and improve.

Lessons from the real world loop back to the requirements for the next version.

Part A

A design method you can reuse

Turning the five layers into a series of questions.

STEP 1 · THE PROBLEM AND THE PEOPLE

“Our greenhouse loses about 10% of its seedlings each summer because nobody notices when it overheats at weekends.”

Ask: what problem? Who uses it? What is it worth?

Eight questions to ask

Question	Why it matters
What must we measure or control?	Chooses the sensors and actuators
How often, and how accurately?	Battery life, data volume, sensor choice
Where will the devices be?	Connectivity, enclosures and power
How are they powered?	Often the hardest constraint of all
How fast must it react?	What runs at the edge or in the cloud
How many devices, now and later?	Cost, management and platform choice
What’s the budget?	Rules options in or out quickly
What data is personal or sensitive?	Shapes security and privacy design

STEP 3 • DESIGN EACH LAYER

Choose each layer — and write down why



Devices

Which sensors, actuators and chip?



Connectivity

Which radio? Which protocol?



Edge

What must be decided locally?



Cloud platform

Where is data kept and analysed?



Application

Who sees what, on which screen?



Security

Applies across every layer.

Every choice traces back to a requirement

Layer	Choice	Why
Devices	Soil and temperature sensors; relay for the valve	Measures what matters; switches the valve safely
Connectivity	LoRaWAN to one farmhouse gateway	Kilometres of range, years on a battery
Edge	Open below 30% moisture, close above 45%	Works if the internet drops; no flickering
Cloud	Store readings; alert if a sensor is silent for 2 h	Checks across all fields, plus history
Application	Phone dashboard: one status tile per field	See what needs attention in seconds
Security	Own keys per sensor; MFA; signed updates	Hard to fake readings; can be patched

The smart irrigation station

End to end: a smart irrigation station

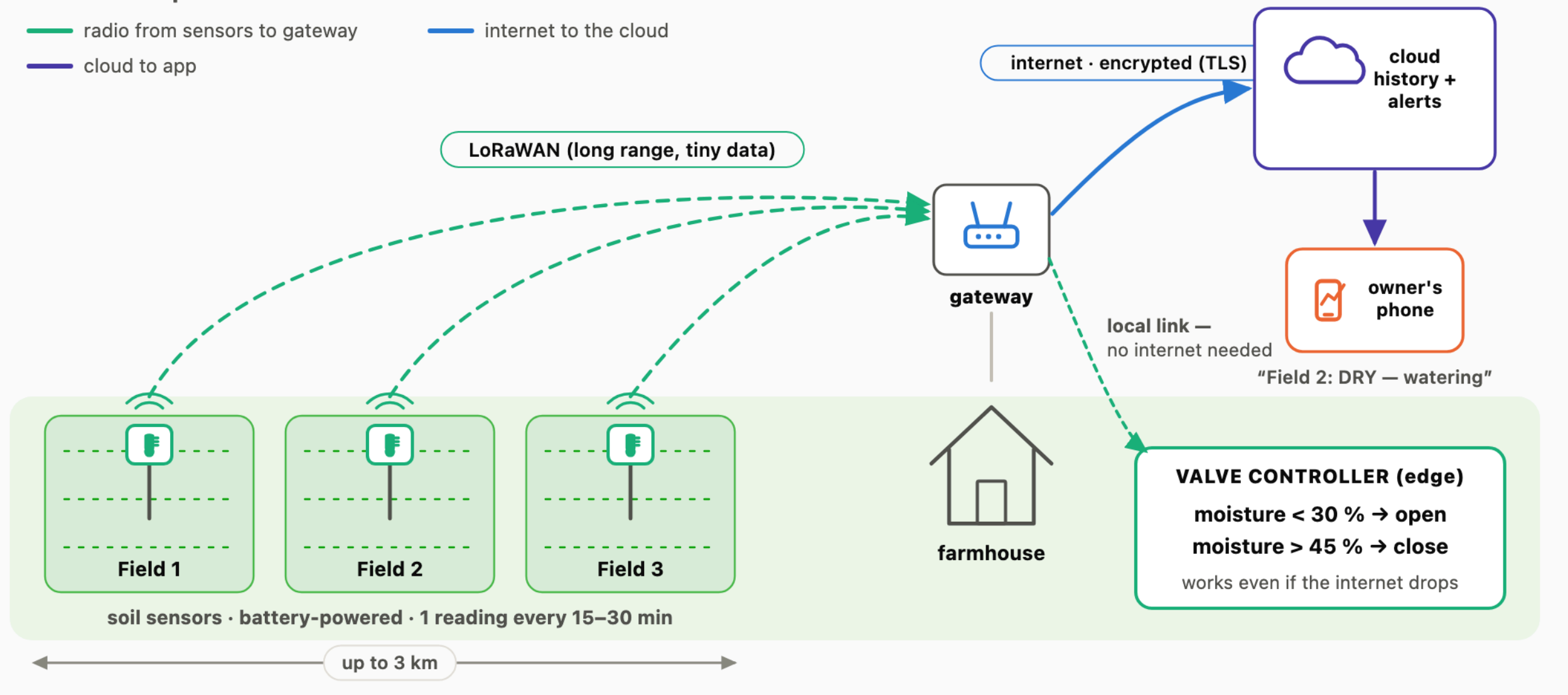
Every choice traces back to a requirement: far away, no mains power, tiny data, must work offline.

What each hop carries:

— radio from sensors to gateway

— internet to the cloud

— cloud to app



PAUSE AND CHECK

Why is the “open the valve” rule at the edge, while the “sensor has gone silent” rule is in the cloud?

Think about what happens when the internet is down. Talk to your neighbour for one minute.

Part B

Case studies from real industries

Same five layers every time. Watch why each choice suits its environment.

CASE STUDY 1

Smart home automation



Connectivity

Wi-Fi for cameras;
Zigbee or Thread for
battery sensors.



Matter

Devices from
different brands
work together.



Edge

A local hub keeps
rules running offline.



Challenges

Privacy, easy setup,
many brands.



CASE STUDY 2

Industrial equipment monitoring

- Vibration, heat and current sensors on motors
- **Edge gateways** analyse data on site
- Cloud predicts failures across the factory
- Downtime cut by **30–50%** (McKinsey)
- Old machines, harsh sites, strict security



CASE STUDY 3

Smart agriculture

- Soil, weather and tank-level sensors
- **LoRaWAN** or **NB-IoT**: big farms, small messages
- Irrigation decided locally; cloud adds forecasts
- Challenges: power, weather, rural signal, cost

WHAT THE CASE STUDIES HAVE IN COMMON

Same five layers, different choices

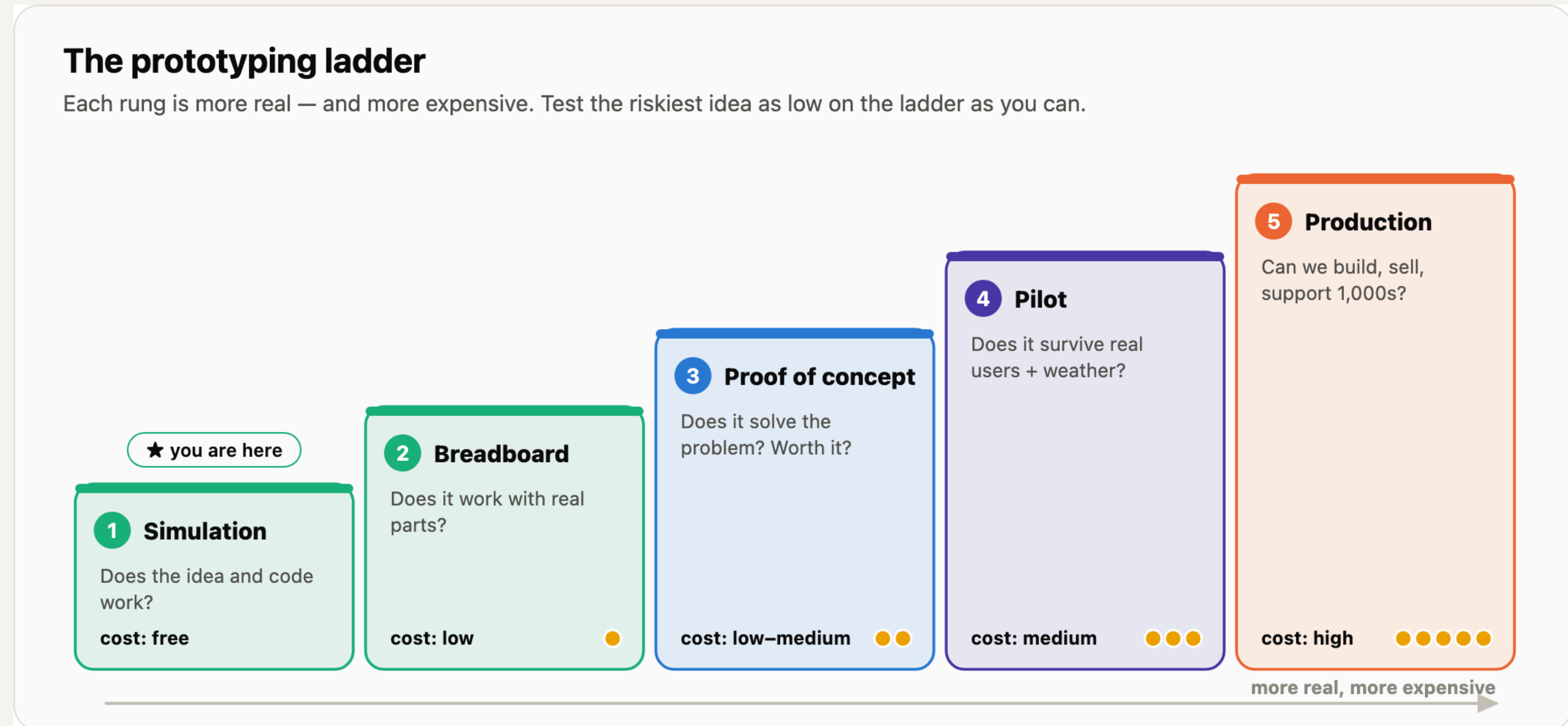
	Smart home	Industry	Agriculture
Main goal	Comfort, security, energy	Avoid breakdowns	Save water and labour
Connectivity	Wi-Fi, Zigbee/Thread, BLE	Wired + gateways	LoRaWAN, NB-IoT
At the edge	Local hub rules	Vibration analysis	Irrigation control
In the cloud	Remote access, history	Cross-factory prediction	Weather-aware planning
Biggest challenge	Privacy, ease of use	Old machines, safety	Power, remote places

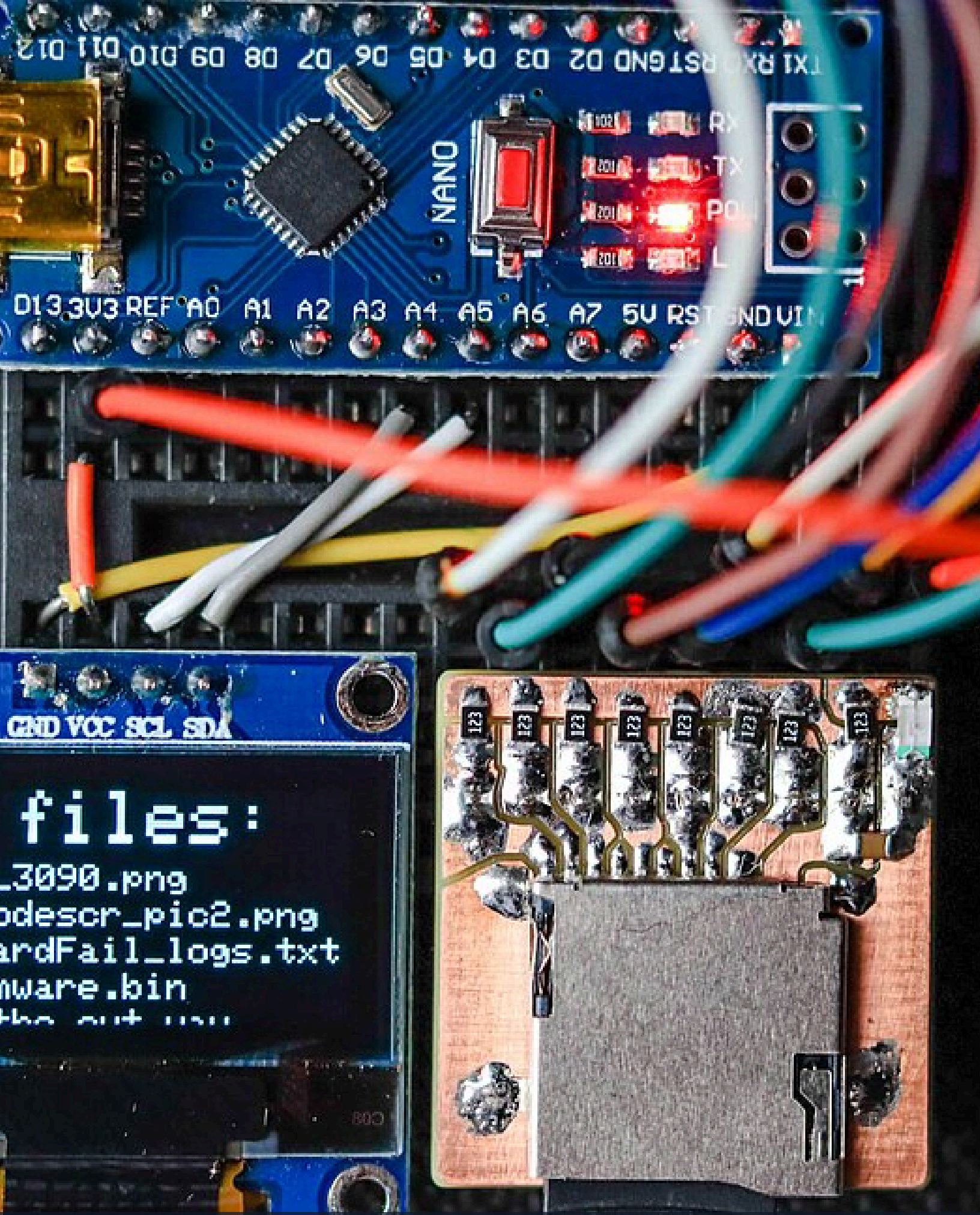
Part C

Prototyping, then the real world

Test ideas cheaply first. Then face the problems the lab never shows you.

The prototyping ladder





PROOF OF CONCEPT

What a proof of concept must prove

- **Can it be done?** Range, battery, accuracy
- **Is it worth doing?** Savings, and real users
- Works but saves nothing? Still a failed PoC
- Test your **riskiest idea** first

Challenges out in the field



Power

Deep sleep;
measure real
battery life.



Connectivity

Store readings
offline, then send
them later.



Environment

Heat and humidity:
use a strong case.



Data quality

Sensors drift:
calibrate them.

Challenges of running at scale



Scale

Set up and update thousands.



Standards

Open standards avoid lock-in.



Total cost

Fees, visits and batteries, for years.



Security & law

Certified radios; laws like PDPA.



People

Involve and train users early.

Memory aid: **power, signal, scale, weather, data, standards, cost, security, people.**

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *If the prototype works, the product is almost done.*

✓ Fact A prototype is often less than half the journey.

✗ Myth *Design the perfect system before building anything.*

✓ Fact Real teams learn by building cheap prototypes.

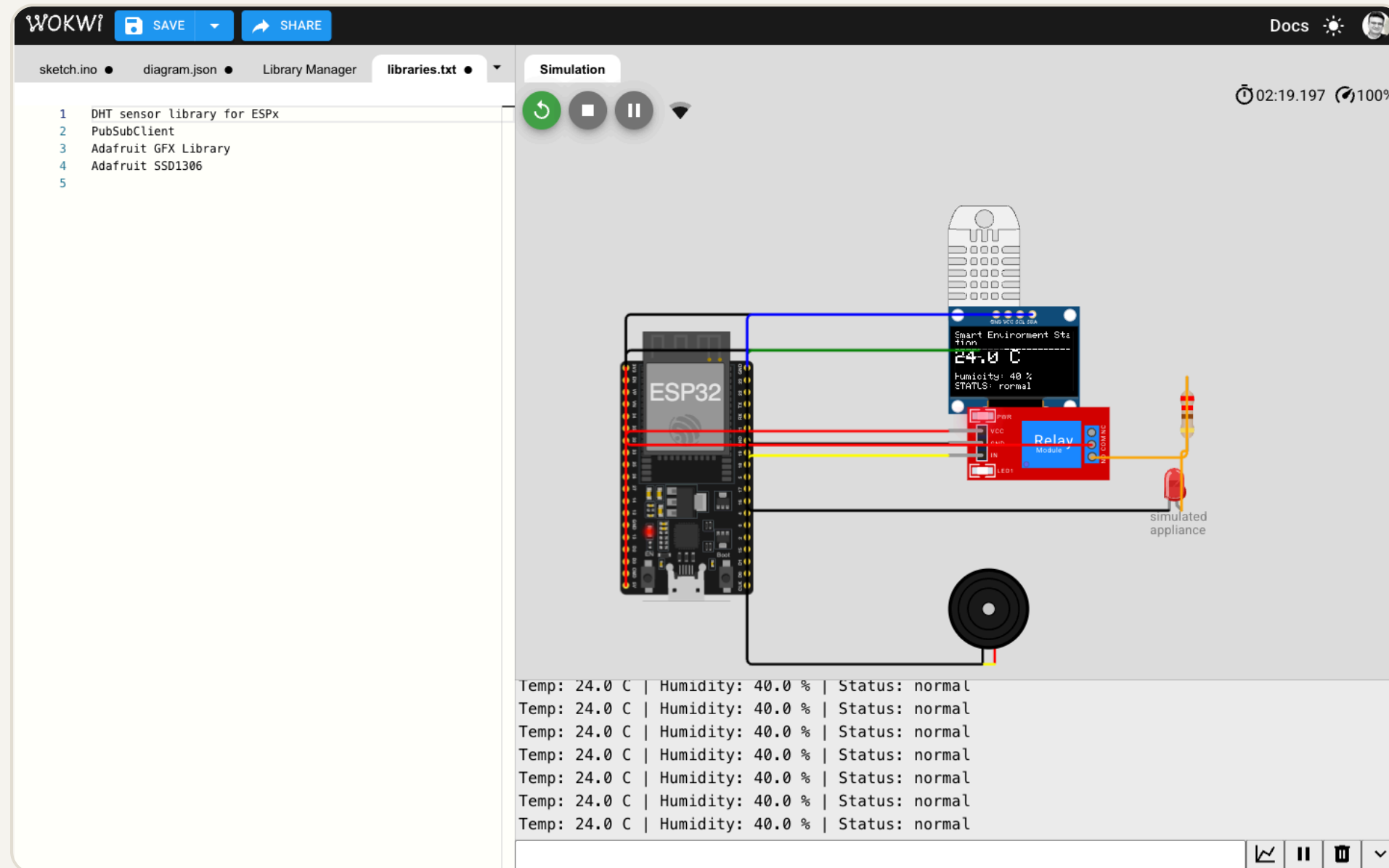
✗ Myth *The most advanced technology gives the best result.*

✓ Fact The simplest design that meets the needs.

✗ Myth *Once devices are installed, the project is done.*

✓ Fact Deployment is the start of years of care.

Capstone: Smart Environment Station



- 1 Open a new ESP32 project in Wokwi
- 2 Paste sketch.ino, and change **YOUR_NAME**
- 3 Paste diagram.json, then add libraries.txt
- 4 Press Play. The first build is slow: be patient
- 5 Push the DHT22 above **28 °C**: watch it react



Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

PROJECT SHOWCASE

Present your project in two minutes



1. The problem

“This station protects a room from overheating.”



2. The layers

Point to devices, connectivity, edge, cloud, app.



3. The demo

Above 28 °C: OLED alarm, relay, buzzer, MQTT.



4. One fix

What would you change before trusting it?

Show it to someone next to you, or record a short video.

KEY TAKEAWAYS

What to remember

- ✓ Start with **the problem and the people**, then requirements
- ✓ Every good design choice **traces back to a requirement**
- ✓ Homes, industry, farms: **same five layers**, different choices
- ✓ Prototype up the ladder; test the **riskiest idea** first
- ✓ A PoC must be both **possible** and **worth doing**
- ✓ Plan for power, signal, scale, weather, data, cost... and **people**

Quick quiz

Q1 Why should an IoT design start with the problem, not a technology?

Q2 In the irrigation example, why LoRaWAN instead of Wi-Fi?

Q3 Put in order: pilot, simulation, production, proof of concept, breadboard.

Q4 Sensors work; the farm saves no water. Did the PoC succeed?

THANK YOU

You built a real IoT system. Keep going!

Next steps: keep experimenting in Wokwi, try real hardware, secure your project, explore Node-RED or Home Assistant, and go deeper on one layer.



Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop